BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 080 · Lesson 080 · Insertion Sort: Grow the Sorted Prefix

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What gets fixed after a bubble pass?

Q2. What does insertion-sort key mean?

Q3. What does sorted prefix mean?

Q4. BST root10 with right child15: insert key12. Show comparisons and position.

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I Do · Read new knowledge

Key vocabulary

insertion sort — Insert key into ordered prefix.

key — Value currently being inserted.

shift — Move existing value to make room.

sorted prefix — Already-sorted left section.

New learning

Understand — The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

Trace — Shift larger elements one position to the right, then insert key in the vacant position.

Explain — Worst $O(n^2)$, already-sorted best O(n), auxiliary O(1); especially useful on small/nearly sorted arrays.

Worked example

Insertion-sort [4,1,3]; give states after each key.

1. Check the working — Key1→[1,4,3]; key3→[1,3,4].

We Do · Practise with guidance

Insert key2 in sorted prefix [3,5] of [3,5,2].

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Shift5 and3 right then write2 at index0, giving [2,3,5].
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Check the working

Shift5 and3 right then write2 at index0, giving [2,3,5].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give arrays after first two key insertions into [5,2,4,1]. [6]

Question 2

Complete insertion of final key1 into [2,4,5,1], showing working. [6]

Question 3

Compare insertion and bubble sort method/time on already sorted input. [6]

Exit Ticket · Answer before checking

Insertion sort always requires a separate second array to store sorted values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Largest remaining value at end.

Q2. Item currently being placed.

Q3. Initial already-ordered portion.

Q4. 12>10 so go right;12<15 so insert as left child of15.

You Do pages · Independent questions

Q1. [2,5,4,1] [3]; [2,4,5,1] [3].

Q2. Shift5 [1], shift4 [1], shift2 [1], insert1 at0 [1], final [1,2,4,5] [2].

Q3. Insertion compares backward and shifts [2], O(n) best case [1]; bubble adjacent passes/swaps [2], early-stop optimisation needed for O(n) best [1].

Exit Ticket · Model answer

False. It can work in place with constant extra space.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: